

Requirements Phase

Solution Requirements Document

Project Name	Smart Library Request Workflow using ServiceNow
Date	3 July 2026

1. Functional Requirements

FRNo	Functional Requirement (Epic)	SubRequirement / UserStory	Priority
FR-1	Book Management	Librarians can add, update, and maintain book records with details (Title, Author, ISBN, Edition, Status).	High
FR-2	Book Catalog	Students can search and browse the book catalog with real-time availability status (Available, Issued, Lost, Returned).	High
FR-3	Borrow Requests	Students can create and submit borrow requests through the Service Portal with book details and desired date.	High
FR-4	Borrow Requests	Students can view and track the status of their own requests in real-time.	High
FR-5	Librarian Approval	Librarians can review pending borrow requests in a dedicated queue/dashboard.	High
FR-6	Librarian Approval	Librarians can approve or reject borrow requests with optional comments/reasons.	High
FR-7	Workflow Automation	Book status automatically updates to 'Issued' when a request is approved by librarian.	High
FR-8	Workflow Automation	Book status automatically updates to 'Available' when a book is returned.	High

FR-9	BookReturn	Librarians can record book returns and update request status to 'Returned' or 'Completed'.	High
FR-10	Role-Based Access	Students can only access and modify their own requests; librarians can access all requests and book records.	High
FR-11	Role-Based Access	Implement ACLs (Access Control Lists) for Student, Librarian, and Administrator roles.	High
FR-12	Email Notifications	Send email notification to student when request is approved with book availability.	Medium
FR-13	Email Notifications	Send email notification to student when request is rejected with reason.	Medium
FR-14	Email Notifications	Send reminder email to student before due date to return borrowed book.	Medium
FR-15	Reporting	Generate 'Most Borrowed Books' report showing top 5-10 most requested books.	Medium
FR-16	Reporting	Generate 'Request Approval Rate' report showing approval vs rejection statistics.	Medium
FR-17	Reporting	Generate 'Overdue Books' report tracking books not returned by due date.	Low
FR-18	Dashboard	Create librarian dashboard showing pending requests, approval queue, and SLA compliance.	Medium
FR-19	Dashboard	Create student dashboard showing personal request history and status.	Medium
FR-20	Validation Rules	Prevent duplicate requests for same book by same student.	Medium

2. Non-Functional Requirements

NFRNo	Non-FunctionalRequirement	Description
NFR-1	Usability	Simple, intuitive Service Portal interface for students and librarians with minimal training required.
NFR-2	UserExperience (UX)	Mobile-responsive design; accessible from desktop and mobile devices via ServiceNow portal.
NFR-3	Security	Role-based Access Control (RBAC) with ACLs for Students, Librarians, and Administrators.
NFR-4	DataProtection	Student data and request details protected; audit logs for all approval and status changes.
NFR-5	Reliability	System uptime >99%; automated workflows with error handling and retry mechanisms.
NFR-6	Performance	Request submission, approval, and book status updates complete within <2 seconds.
NFR-7	Availability	Accessible 24/7 through ServiceNow platform; no downtime except scheduled maintenance.
NFR-8	Scalability	System designed to support multiple libraries, additional resource types, and future campus expansion.
NFR-9	DataIntegrity	Real-time book status updates; no duplicate records; data consistency across all tables.
NFR-10	Maintainability	Clear documentation, modular design, and code standards for easy future enhancements.
NFR-11	Compliance	Adherence to institutional policies and FERPA/GDPR requirements for student data.
NFR-12	Integration	Compatible with existing ServiceNow instance and can integrate with future systems.

3. Technical Requirements

Component	Requirement	Details
Custom Tables	Book Table (u_book)	Fields: Title, Author, ISBN, Edition, Status, Available Copies, Location
Custom Tables	Borrow Request Table (u_borrow_request)	Fields: Requested By, Book (Reference), Request Date, Status, Approval Date
Service Catalog	Catalog Item for Borrow Request	Multi-step form with book selection, validation, and auto-filled student details
Workflow Automation	Flow Designer for Auto-Status Update	Trigger: Request Status = Approved → Action: Update Book Status=Issued
Workflow Automation	Flow Designer for Return Processing	Trigger: Book Returned → Action: Update Book Status=Available
Access Control	ACLs for Tables	Book: Librarians = CRUD, Students = Read. Borrow Request: Students = Create/Readonly, Librarians = All
Notifications	Email Notifications	Automated emails on approval, rejection, and due date reminders
Reporting	Reports & Dashboards	Most Borrowed Books, Request Approval Rate, Librarian Dashboard, Student Dashboard
Database	Data Storage	ServiceNow database with normalized schema for books and borrow requests

4. Constraints & Assumptions

Constraints

- System must operate within existing ServiceNow instance; no additional platforms or tools.
- Barcode/RFID scanning not in scope; manual status updates by librarians for returns.
- Fine calculation for overdue books deferred to future phase.
- Multi-library support planned for future; initially single library only.

- Integration with external library systems (ISBN databases) out of scope.
- Mobile app development out of scope; web portal only via ServiceNow.

Assumptions

- All users have access to ServiceNow instance and basic IT skills.
- Book inventory will be fully populated in the system by go-live.
- Librarians will actively monitor and process requests within SLA timeframes.
- Email notifications will be delivered successfully; users have valid email addresses.
- No downtime for the library during system implementation.
- ServiceNow instance remains stable and properly maintained.